Statistical Report

Determinants of Belgian Household Energy Consumption

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1 Background

The goal of "*Carbon Neutrality*" is gaining increasing momentum in the European political arena amid substantial indicators of an incoming climate change which might have deleterious effects at the societal, economic and, most importantly, environmental level. Household energy consumption plays a pivotal role in overall social sustainability since it accounts for a substantial share of the total energy demand in most developed countries. Within this sector, kilowatt-hour (kWh) consumption provides a useful measurement for overall household energy demand and can be put into relationship to structural characteristics of the buildings as well as the socio-economic profile of their occupants.

Dwelling related attributes such as floor area, dwelling type, construction period, and Energy Performance Certificate (EPC) labels are consistently identified as important predictors of electricity demand (McLoughlin et al., 2013; Santin et al., 2009). Intuitively, larger dwellings typically require more energy for lighting, heating, cooling, and appliance use, while older buildings are constructed with materials which provides poorer insulation and therefore larger heat dissipation. Nevertheless, these variables interacts with socio-economic factors. Larger households topically host a higher number of occupants, therefore they can share the consumption for heating, refrigerators and air conditioning and, consequently, lowering the *per capita* consumption. Similarly, one might speculate that high-class EPC label buildings are mostly prerogative of high-income classes which might use a larger number of appliances and indulge into energy-consuming behaviours going to erode the benefits of a properly insulated house in terms of absolute energy savings (Aydin et al., 2017). This cross-sectional study collected data from 2500 households in Belgium and examined the relationship between households annual kWh consumption and several predictors using a linear regression model.

2 Data management

2.1 Data Cleaning

Three observations (rows 66, 36, and 135) were removed from the dataset according to the protocol The dataset was subsequently restricted to complete cases by excluding all observations (rows) with missing values, allowing estimation of the regression β coefficients via the least squares method. This approach relies on the assumption that the data are missing at random (MAR). None of the variables displayed missingness above 5% (see table 4). Five observations in the `annual_kwh` (rows 267, 1330, 1570, 369, 12) variable displayed values two orders of magnitude larger than the others (172 to 312 times larger) despite being comparable to other records across the other variables (indoor surface, EPC labe, etc. . .). Therefore, these were also removed as were considered implausible and likely attributed to imputation error (see section 6.2 for details).

Categorical variables were encoded as factors, and reference levels were specified. Namely, the reference levels were defined as *"Pre-1970"* for `building_era`, *"A"* for `epc_label`, *"No"* for `pet_ownership` and *"Red"* for `roof_color`, against which all other categories were compared.

3 Descriptive Statistics

3.1 Categorical variables

The distribution of EPC labels is relatively balanced across categories. Labels A (22.2%) and C (21.3%) are the most common, followed by B (17.6%) and D (15.7%). Lower-efficiency categories E occur less frequently, 13.6% and F 9.6% respectively. Buildings constructed between 1970 and 2000 represent 34.4% of the sample, pre-1970 33.1%, and post-2000 32.6%. Occupancy count ranges from 1 to 7 with all counts accounting for approximately 14-15%. A breakdown is provided in table 2. None of the categories appears too sparse, suggesting that all variables can be retained for modelling without concerns about low cell counts. From the exploratory analysis, three categorical variables seem to correlate with annual Kwh consumption: EPC label, building era and and occupancy count (figures 4-5). The other categorical variables do not display an apparent strong correlation with electricity consumption with similar means across each level. Moreover, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras (figure 7).

3.2 Continuous variables

On average, households consume 5,874.76 kWh annually (considering the 5 outliers removed), with a relatively large standard deviation (2224.25 kWh). The median consumption (5432.07 kWh) is slightly lower than the mean, suggesting a right-skewed distribution, which is also supported by the QQ plot showing upward deviations above the diagonal line in the upper tail (figure 3). Floor area averages 158.25 m^2 with a standard deviation of 36.50 m^2, and a range from 60.26 m^2 to 341.30 m^2. Household income averages €48842.90, with a standard deviation of €22534.97 and a range from €8487.91 to €206367.91. As expected for this type of variable the QQ plot shows strong upper-tail deviation, suggesting left skewness driven by high-income outliers. A breakdown for each variable is provided in table 3.

The correlation matrix reveals that annual electricity consumption has a moderate positive correlation with floor area ($r = 0.38$) and with income ($r = 0.29$), suggesting that larger apartments and wealthier households tend to consume more energy. Importantly, two predictors exhibit a strong positive correlation with each other, namely floor area and income ($r = 0.72$). Controlling for the income level, the EPC label distribution appears similar across income quantiles (figure 8) and comparable to the overall EPC distribution in the aggregate dataset. Indoor floor area does not associate with the size of the household since median floor area and interquartile ranges remain relatively stable when controlling for permanent residents, suggesting that larger households do not systematically live in larger dwellings in this dataset (figure 9). Distance to Brussels shows virtually no correlation with the other variables, suggesting that location relative to Brussels does not predict household energy use, dwelling size, or income in this dataset. More details are provided in sections 6.4 and 6.5 of the appendix.

4 Inferential Statistics

4.1 Regression model

The initial model contained all variables plus interaction terms between income and number of occupants, EPC label and floor area, EPC label and number of occupants, and income and EPC label. Interactions were selected based on literature and protocol research questions. A quadratic term was considered for `occupancy_count` consistently with a possible decreasing marginal effect of additional occupants. We will refer to the initial model as *saturated*.

Given that the primary objective of this study was to identify the set of regressors that best predict energy demand, explanatory variables retention was determined using a backward stepwise procedure based on the Bayesian Information Criterion (BIC) rather then exclusively on p-values. BIC favours more parsimonious models therefore lowering the risk of over-fitting. At each step, the variable whose removal produced the greatest reduction in BIC was eliminated, and the process continued until no further decrease was observed.

The outcome variable `annual_kwh` was initially regressed using row data. Diagnostic plots (fig. 12) showed some violations of the standard assumptions. In particular, the residuals versus fitted plot indicates the presence of heteroskedasticity, with the variance of the residuals increasing for higher fitted values, and mild deviations from normality in the Q-Q plot. The outcome variable `annual_kwh` was then log-transformed resulting in an improvement of the diagnostic plots, particularly by yielding a more symmetric residual distribution and enhancing homoskedasticity. Log-transformed `annual_kwh` was used therafter for all model testing rpocedurs.

The final model retained floor area, EPC label, and the number of occupants as predictors of annual electricity consumption. The interaction terms and other covariates were removed. Nevertheless, `building_era` was retained in the model as it acts as a confounder being related to both `annual_kwh` and `epc_label`. We posit that building era affects energy consumption but also determines the energy calss since EPC label A and B were basically absent before year 2000 (see section 6.4 for details).

Income was not retained in the final model and was not statistically significant (see Table 8). A plausible explanation is that the effect of income on energy consumption operates indirectly through dwelling size. As shown in Table 7, mean energy consumption across income quantiles becomes similar once stratifying by square meter quantiles. Moreover, income and dwelling size are strongly positively correlated. This suggests that dwelling size may act as a mediator in the relationship between income and energy consumption, capturing much of the variability that would otherwise be attributed to income.

Regression results for the saturated model, including p-values, are displayed in table 6. Based on statistical significance only, similar conclusions would have been achieved. Floor area, EPC label, household income, and occupancy count as predictors were the statistically significant ones. No evidence was found for non-linear effects of occupancy or for interaction effects, supporting a parsimonious model as the one selced using BIC criterion.

The results indicate that household size is a strong predictor of energy consumption, with each additional square meter associated with an approximate 0.4% increase in annual consumption. Assuming an average bed room size of 15 squared meters, each extra bedroom consumes on average 6% more energy. Compared to dwelling in energy class A, lower efficiency classes exhibit progressively higher electricity consumption, from about 7% for class B to nearly 48% for class F.

Each additional occupant is responsible for around 16% increase in electricity consumption. No evidence of a decreasing marginal effect of the number of occupants was found at 5% significance level. Since there is no evidence that larger households tend to live in larger dwellings, a plausible explanation for the observed linear relationship between energy consumption and the number of occupants is that energy demand is primarily driven by individual behaviours, such as water use and appliance use, which are largely non-shareable.

The final model achieved a substantially lower BIC (table 7) while maintaining nearly identical R^2 (89%), indicating that almost all the variance in annual consumption is explained by EPC label, house size and number of occupants. Overall, the diagnostic plots indicate that model assumptions are reasonably satisfied, with only minor deviations that are unlikely to substantially affect inference. the residuals vs leverage plot shows a few observations with higher leverage, but none appear to exert undue influence based on Cook's distance, indicating the absence of strongly influential points.

Table 1: Regression results for the final model (p-values are adjusted using the Benjamini-Hochberg procedure)

	Beta	Std. Error	p-value	% Effect	CI 2.5%	CI 97.5%
(Intercept)	7.220	0.020	0.000	136523.834	131245.615	142014.162
sq_meters	0.004	0.000	0.000	0.397	0.383	0.410
epc_labelB	0.087	0.009	0.000	9.094	7.125	11.100
epc_labelC	0.157	0.014	0.000	16.959	13.747	20.262
epc_labelD	0.238	0.015	0.000	26.808	23.155	30.570
epc_labelE	0.316	0.017	0.000	37.150	32.542	41.917
epc_labelF	0.394	0.018	0.000	48.226	43.099	53.538
Built 1970–2000	-0.009	0.009	0.361	-0.856	-2.670	0.991
Built post-2000	-0.010	0.015	0.531	-0.949	-3.860	2.049
occupancy_count	0.151	0.001	0.000	16.311	16.026	16.596

5 Conclusions

The present analysis shows that, as already confirmed by literature, EPC label and number of occupants are the most relevant predictors of annual electricity consumption. Compared to dwelling in energy class A, lower efficiency classes exhibit progressively higher electricity consumption. Building in class F consumes almost 50% more than those in class A. Income does not appear to have a direct effect on energy consumption once dwelling characteristics are accounted for. A possible explanation is that income operates indirectly through larger living spaces (square_meters). There is no evidence that the effect of EPC label varies with household occupancy. Interaction terms between EPC label and occupancy were not retained in the final model and were not statistically significant.

From the policy-maker perspective, data suggest that interventions aimed at reducing residential energy demand should prioritise improving building energy performance. It remains to be established whether encouraging the use of common areas (for example co-working spaces in the building) will impact substantially on total energy demand.

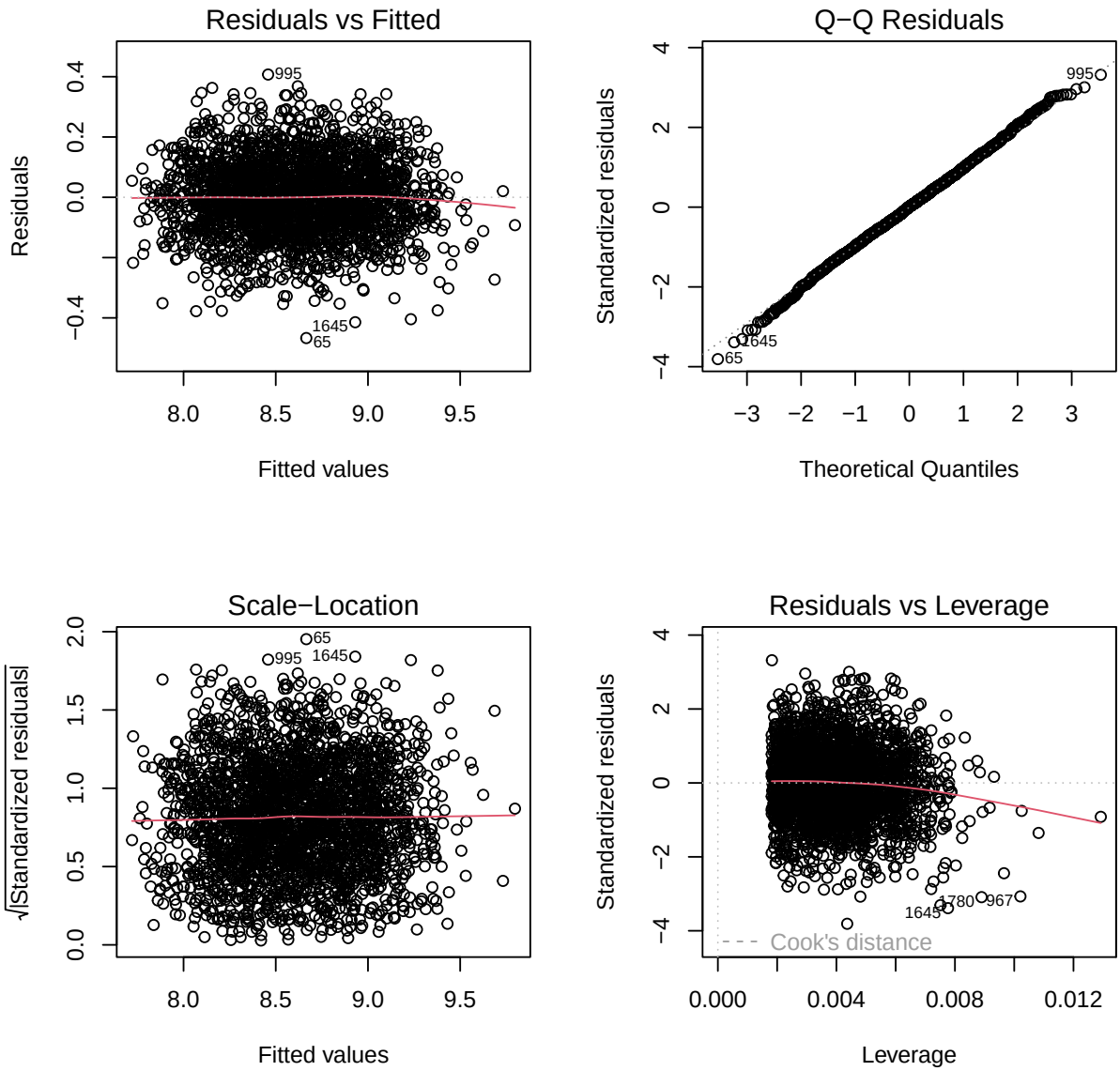


Figure 1: Diagnostic plots for the selected regression model using log-transformed annual energy consuption.

6 Appendix

6.1 Summary tables for non-continuous variables

Table 2: Categorical variables (n, %)

variable	value	n	p
building_era	Pre-1970	820	33.1
building_era	1970-2000	850	34.3
building_era	Post-2000	807	32.6
epc_label	A	551	22.2
epc_label	B	436	17.6
epc_label	C	528	21.3
epc_label	D	387	15.6
epc_label	E	336	13.6
epc_label	F	239	9.6
pet_ownership	0	1496	60.4
pet_ownership	1	981	39.6
roof_color	Red	872	35.2
roof_color	Black	824	33.3
roof_color	Other	781	31.5

6.2 Summary tables for continuous varibales

Table 3: Summary Statistics for Continuous Variables

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	5874.47	158.25	62.91	48842.90
SD	2225.09	36.48	33.04	22534.97
Median	5432.16	155.84	63.17	44535.07
IQR	2991.14	46.96	57.64	25894.30
Min	1824.63	60.26	5.11	8487.91
Max	17197.12	341.30	120.00	206367.91

6.3 Summary tables for missing values

Table 4: Missing data summary

Variable	Missing_n	Missing_pct
annual_kwh	5	0.2
sq_meters	5	0.2
epc_label	5	0.2
income_euro	5	0.2
occupancy_count	5	0.2
dist_to_brussels	5	0.2
building_era	0	0.0
roof_color	0	0.0
pet_ownership	0	0.0

6.4 Summary plots for continuous varibales:exploratory analysis

Figures below refer to the dataset used for the regression model from which missing values and outliers in annual energy consumption were removed as explained in section 2 From the box plot we observe that several variables exhibit right-skewness and upper-tail outliers, particularly income and energy consumption. While annual energy consumption deviates significantly from normality, its log transformed analogue follows an approximate normal distribution dispute a mild deviation from the expected quantiles in the lower and upper tail. The QQ plot for floor area shows only mild departures from normality. The average distance to Brussels is 62.96 km. Its QQ plot indicates noticeable curvature, implying that the distribution deviates from normality, possibly reflecting clustering of households at certain distance ranges during the sampling procedure. Household income QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers.

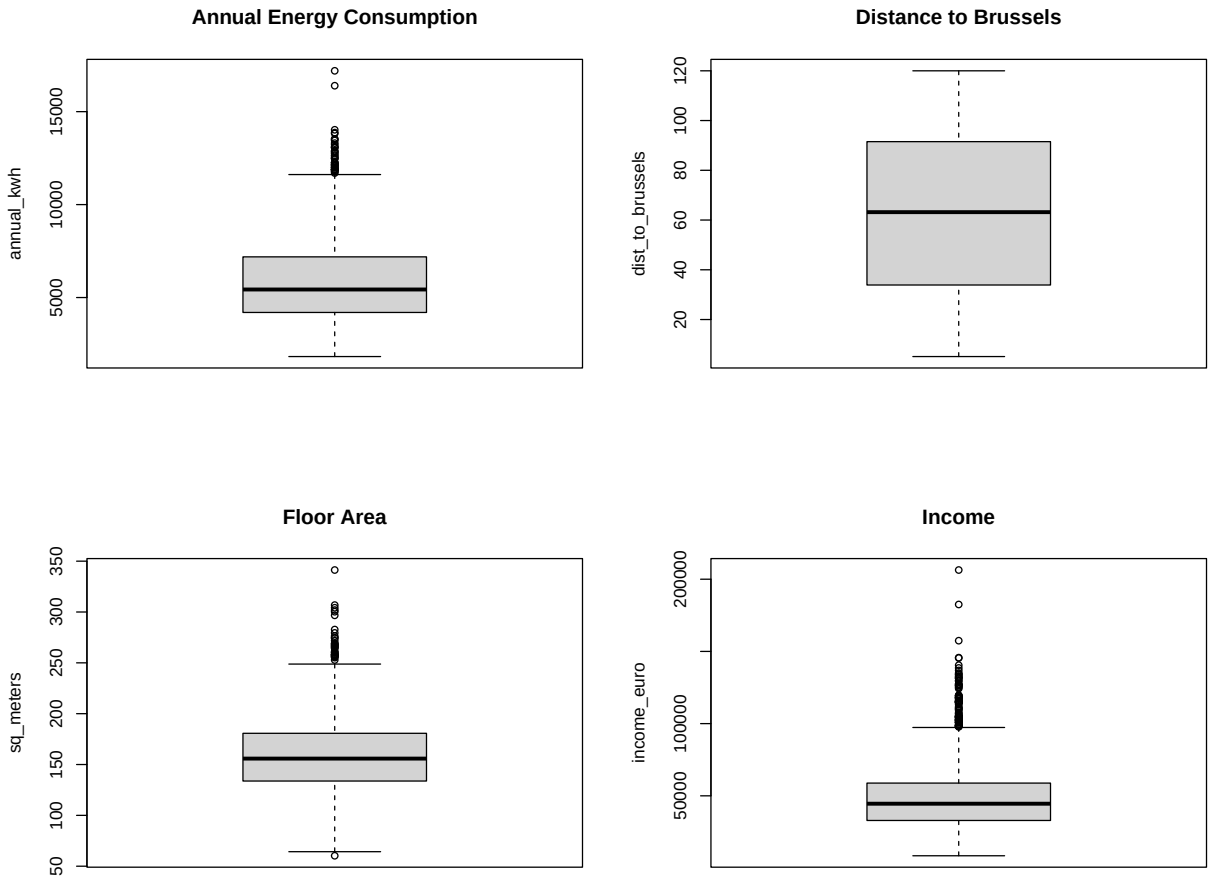


Figure 2: Box plots for continuous variables

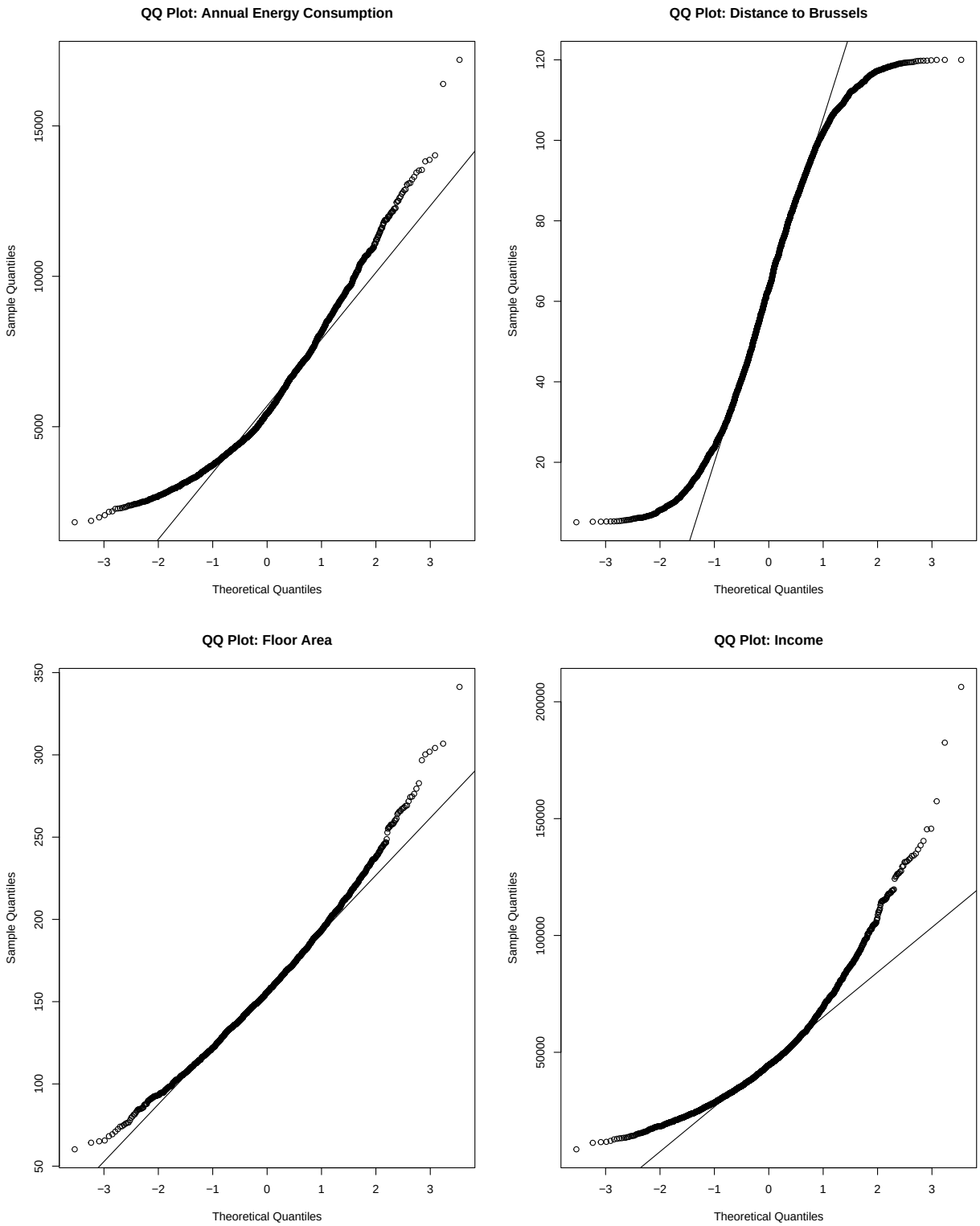


Figure 3: QQ plots for continuous variables

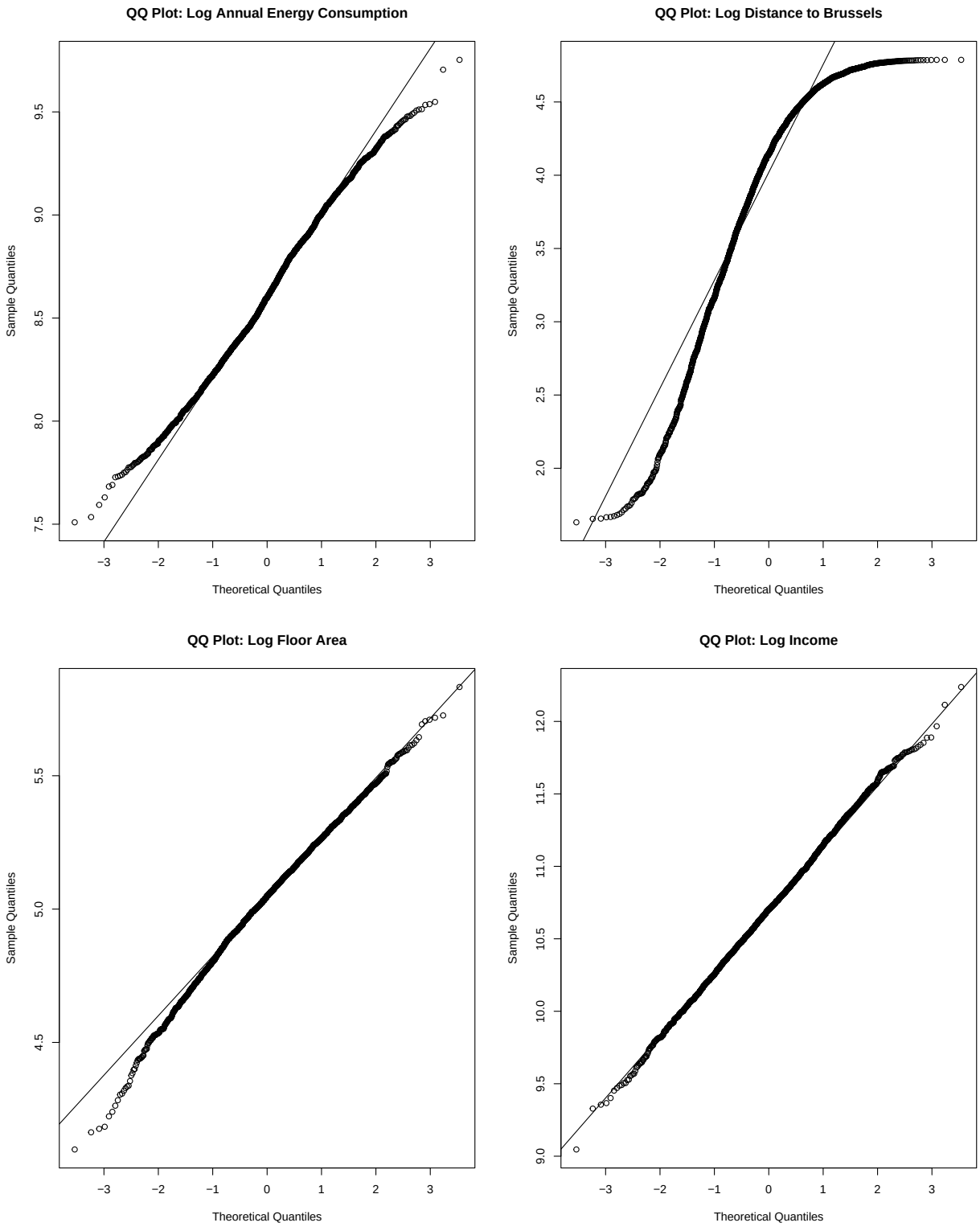
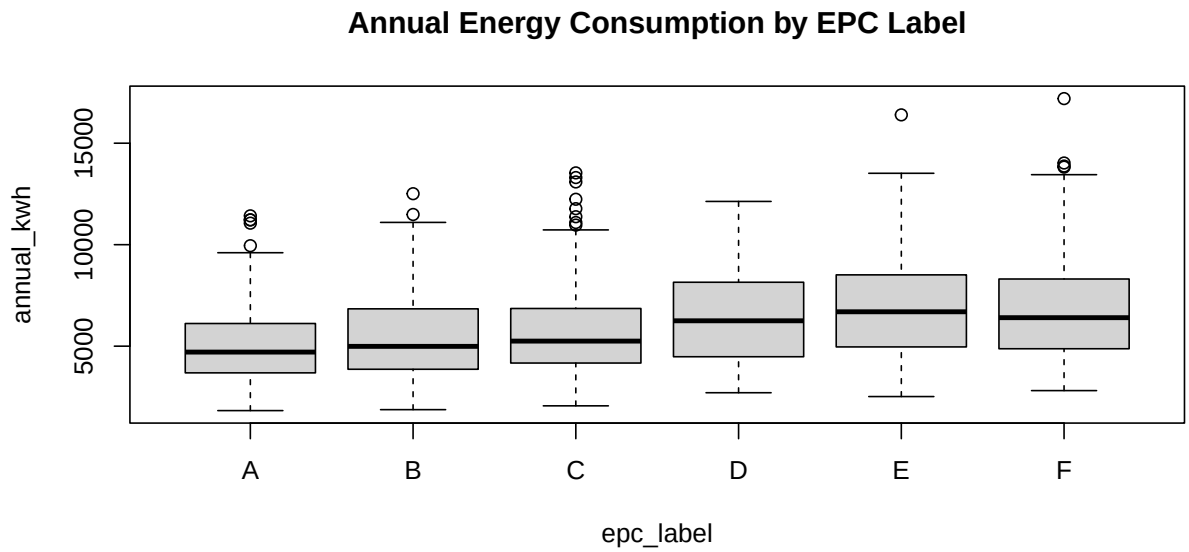


Figure 4: QQ plots for log-transformed continuous variables

6.5 Relationship between annual energy consumption and non-continuous variables: exploratory analysis

Among categorical and ordinal variables, EPC label and occupancy count are the most correlated to annual energy consumption. In addition, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras and most recent buildings. Therefore, building era might play as a confounder in the model.



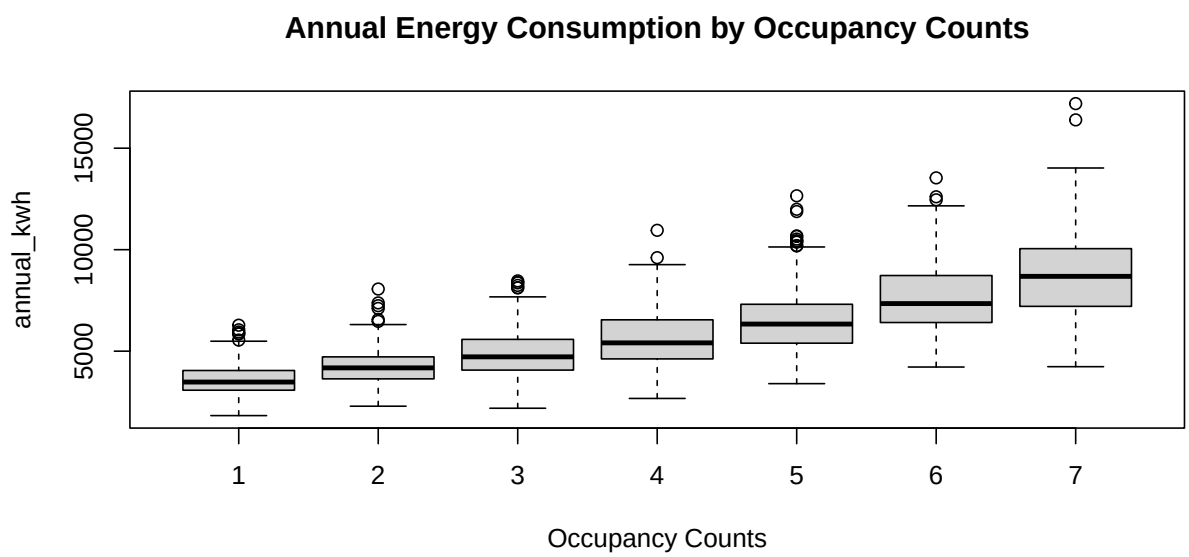
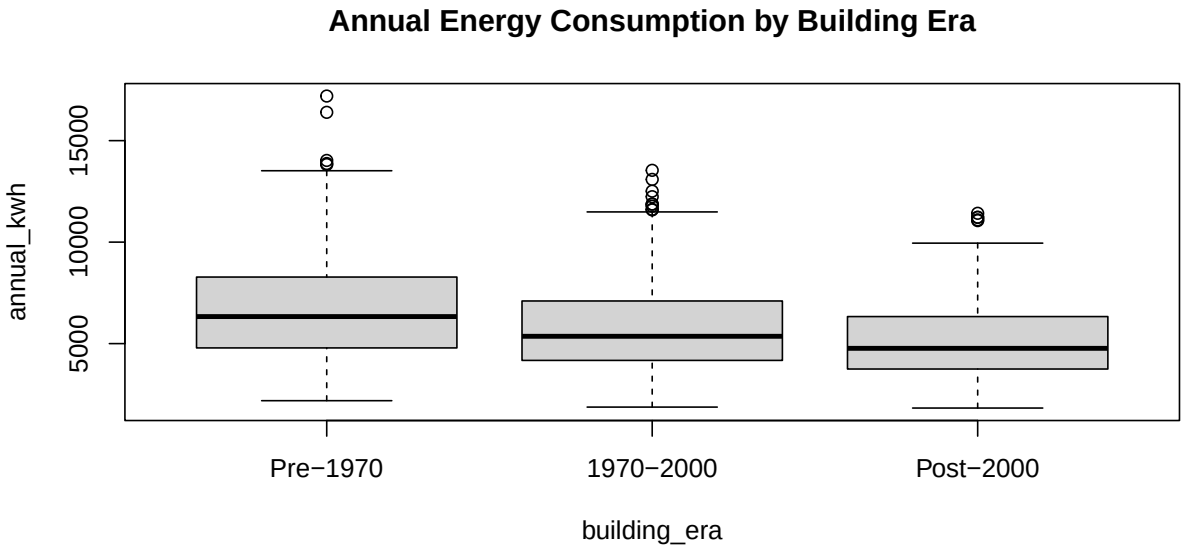


Figure 5: Box plots for annual energy consumption vs categorical variables (part 1)



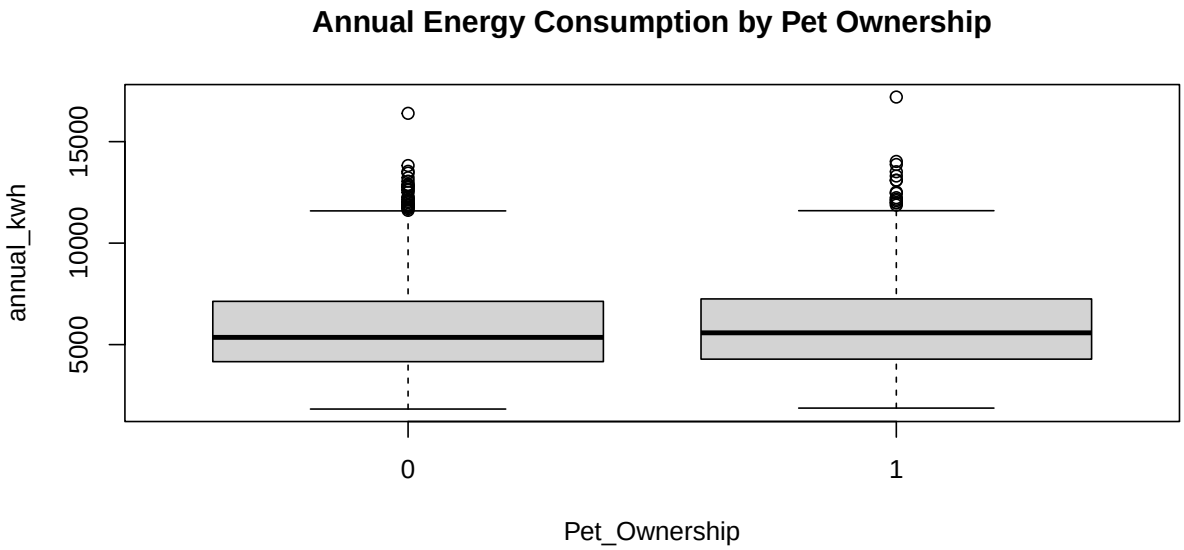


Figure 6: Box plots for annual energy consumption vs categorical variables (part 2)

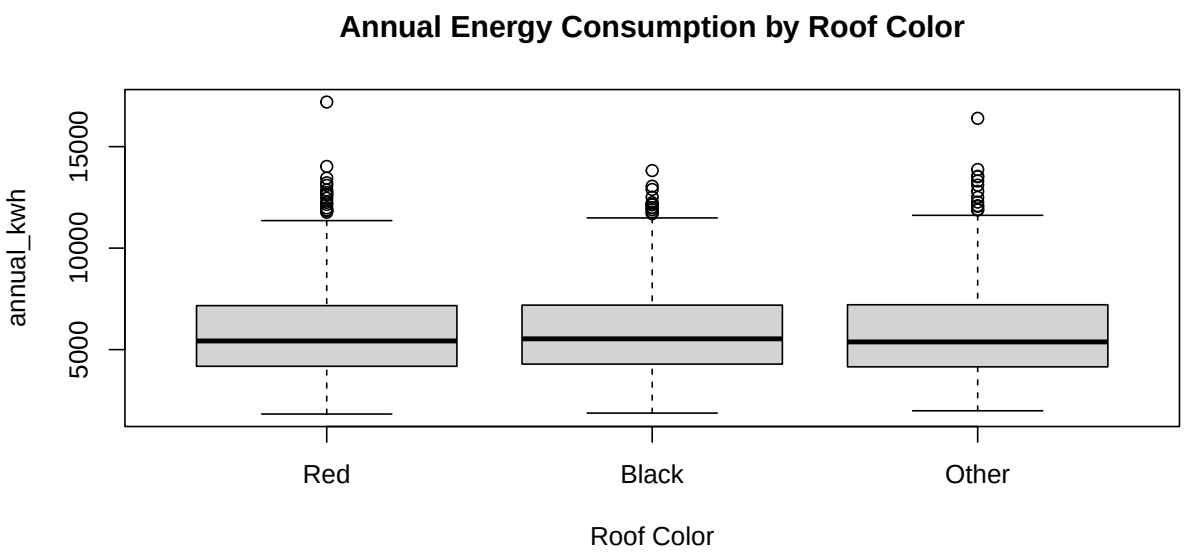
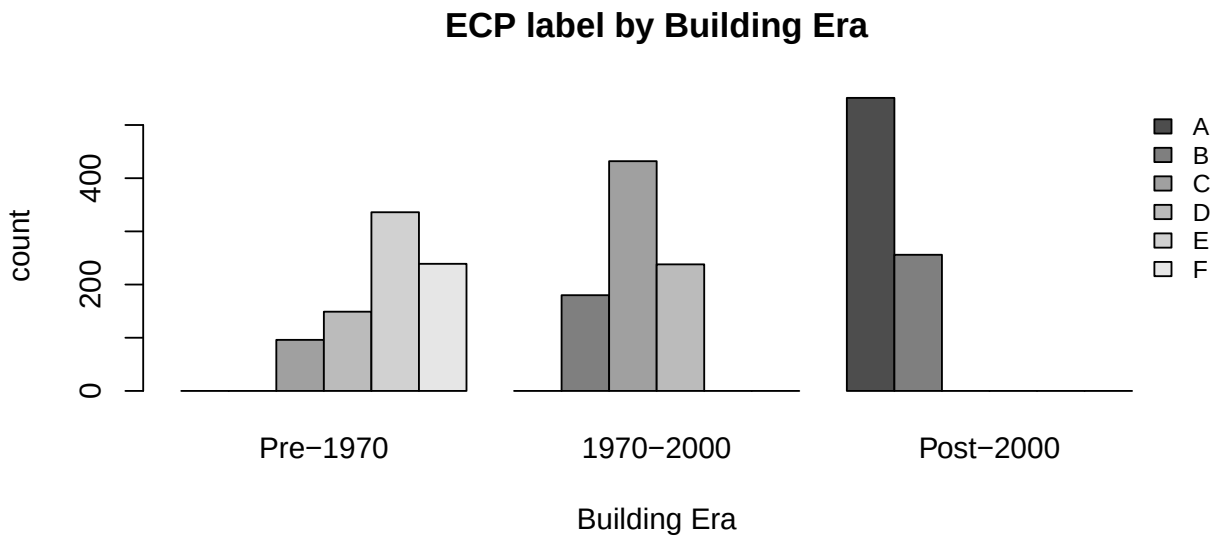


Figure 7: Box plots for annual energy consumption vs categorical variables (part 3)

6.6 Causal relationships among variables

Building era was treated as a confounder in the relationship between EPC label and electricity consumption since it is related to both, the main predictor and the outcome variable.

Income was treated as a mediator in the relationship between EPC electricity consumption and size of the households since people with higher income can afford larger dwellings which in turn consume more electricity. Indeed, the income and apartment size are strongly correlated (rho=0.72).



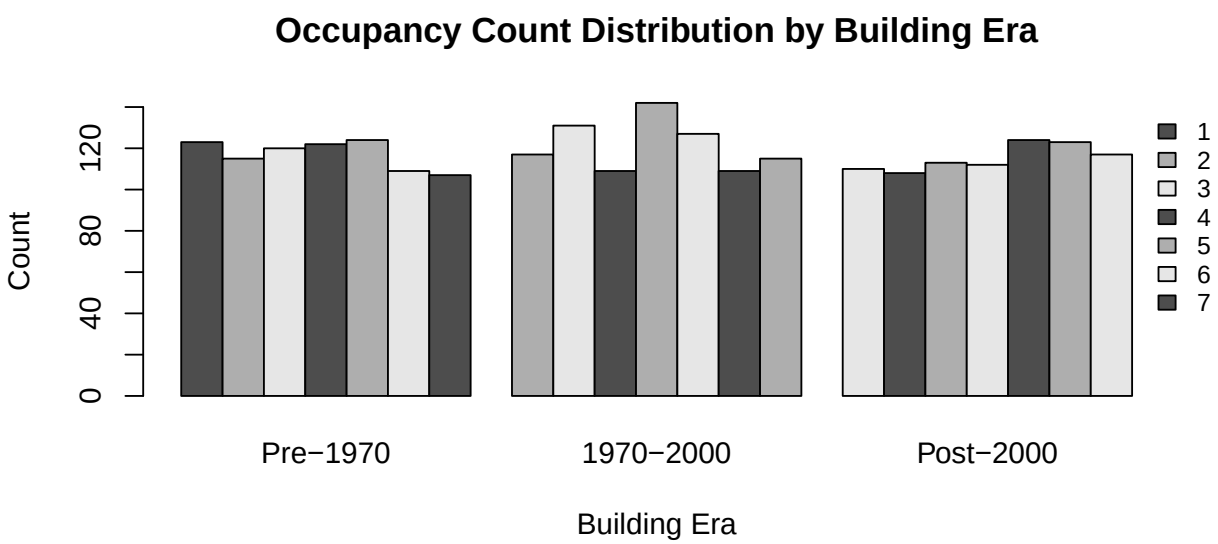


Figure 8: Bar plots predictors against each other (selected cases)

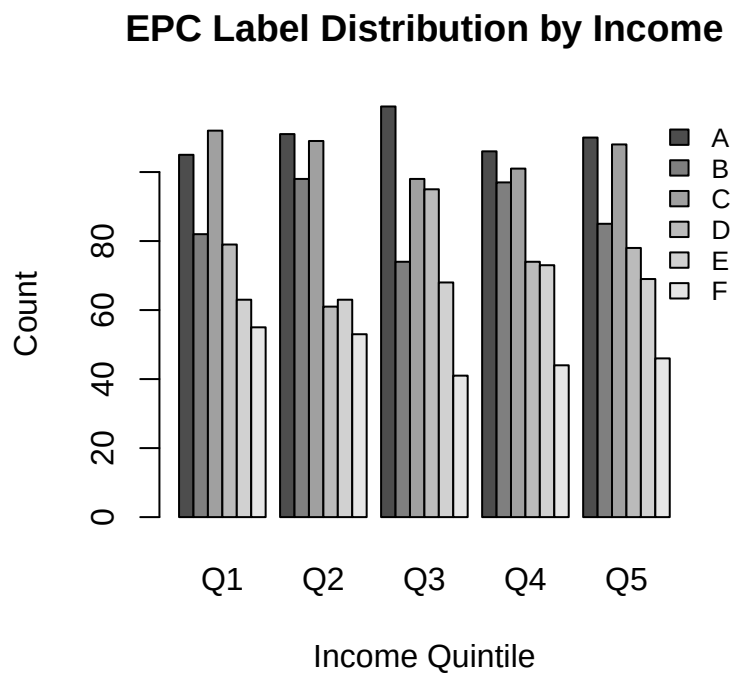


Figure 9: Bar plots of EPC label class by Income percentile

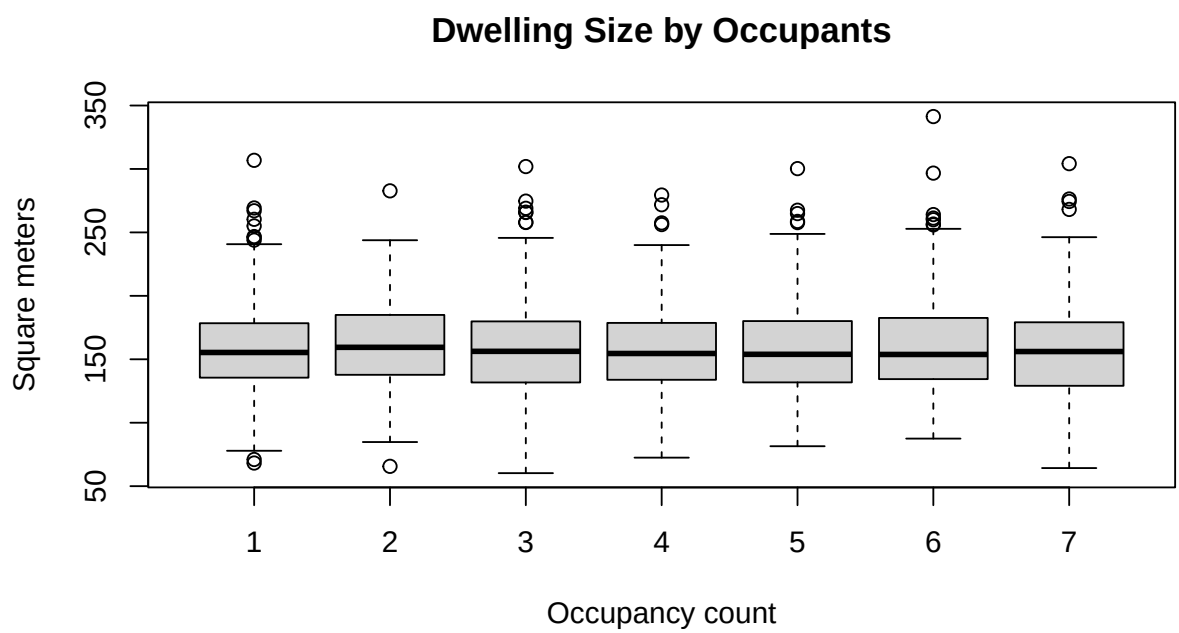


Figure 10: Box plots of dwelings square meters by occupancy count

Table 5: Mean Annual Energy Consumption by Size and Income Quintiles

	Income Quintiles				
	Q1	Q2	Q3	Q4	Q5
Q1	4723	5027	4917	4846	5381
Q2	5287	5459	5165	5747	5610
Q3	5725	5581	5683	5723	6384
Q4	6370	6119	6237	6342	6249
Q5	7866	6782	6079	6883	7323

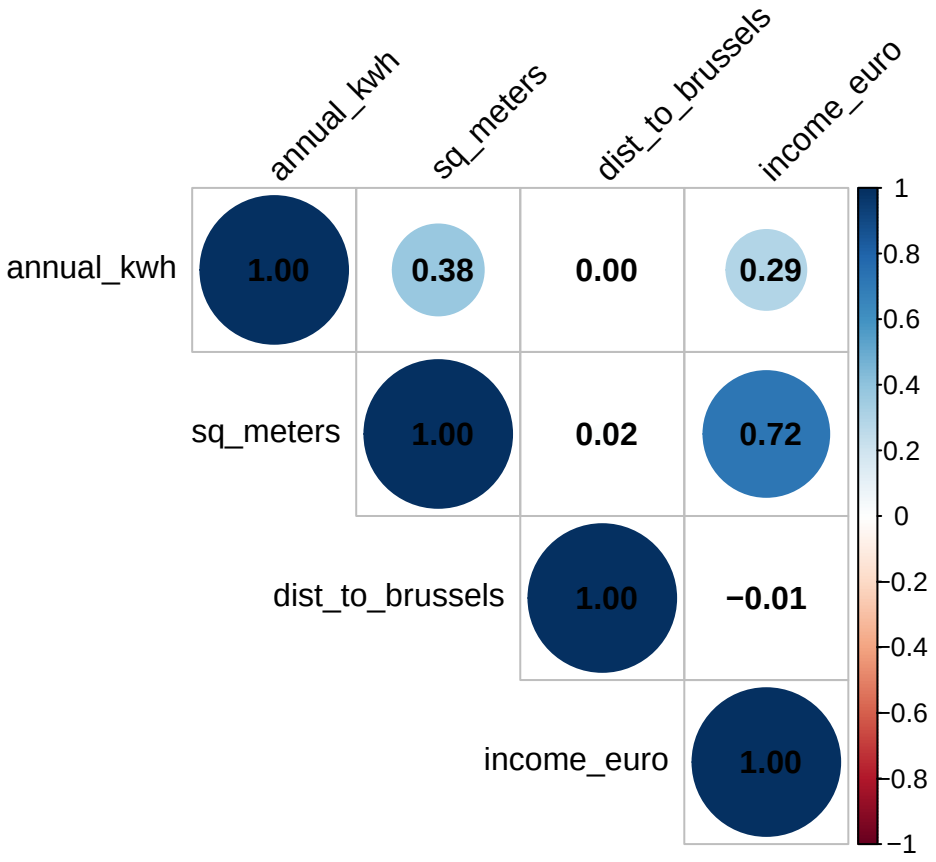


Figure 11: Correlation plots for continuous variables

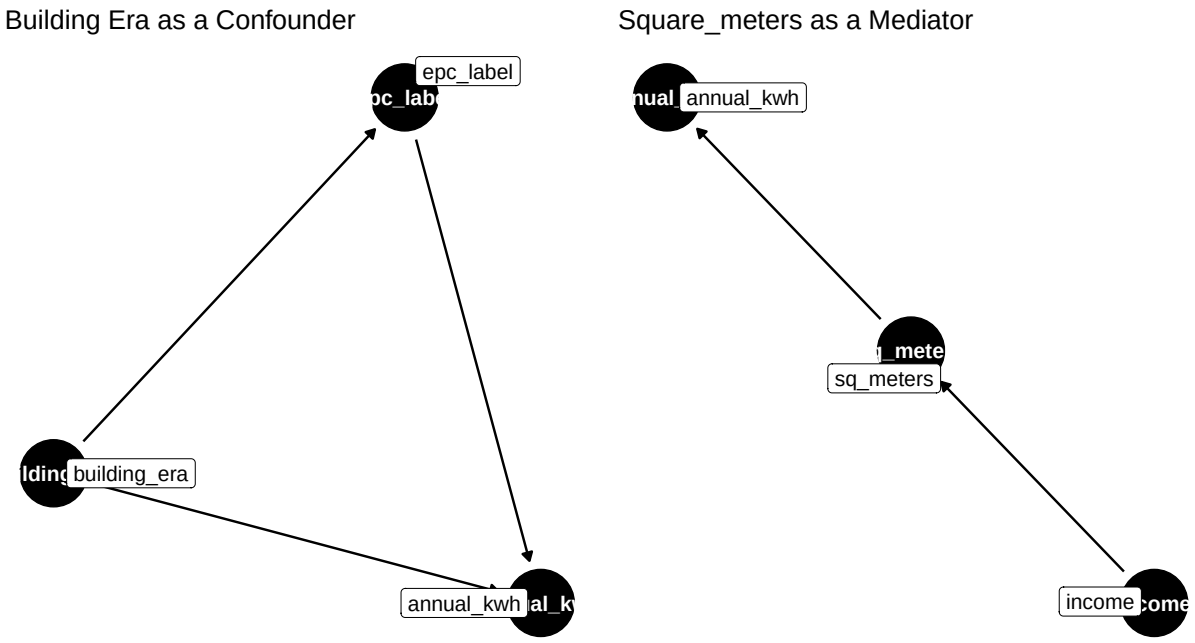


Figure 12: Directed acyclic graphs (DAGs)

6.7 Regression model

The initial regression model was the following:

$$
\begin{aligned}
\text{annual_kwh}_i = \beta_0 &+ \beta_1\,\text{sq_meters}_i + \beta_2\,\text{income_euro}_i + \beta_3\,\text{occupancy_count}_i + \beta_4\,\text{dist_to_brussels}_i \\
&+ \boldsymbol{\beta}_5\,\text{epc_label}_i + \boldsymbol{\beta}_6\,\text{building_era}_i + \boldsymbol{\beta}_7\,\text{roof_color}_i + \beta_8\,\text{pet_ownership}_i \\
&+ \beta_9(\text{income_euro}_i \times \text{occupancy_count}_i) \\
&+ \boldsymbol{\beta}_{10}(\text{epc_label}_i \times \text{sq_meters}_i) \\
&+ \boldsymbol{\beta}_{11}(\text{epc_label}_i \times \text{occupancy_count}_i) \\
&+ \boldsymbol{\beta}_{12}(\text{epc_label}_i \times \text{income_euro}_i) \\
&+ \boldsymbol{\beta}_{13}(\text{occupancy_count}_i^2) + \varepsilon_i
\end{aligned}
$$

The final model retained the following predictors:

$$
\log \text{annual_kwh}_i = \beta_0 + \beta_1\,\text{sq_meters}_i + \beta_2\,\text{occupancy_count}_i + \beta_3\,\text{epc_label}_i + \beta_4\,\text{building_era}_i + \varepsilon_i
$$

6.8 Summary tables of regression results for the saturated model

Table 6: Regression results for the saturated model using log-transformed outcome (p-values are adjusted using the Benjamini-Hochberg procedure).

	Estimate	Std. Error	t value	p-value
(Intercept)	7.05006	0.03897	180.89118	0.00000
sq_meters	0.00501	0.00023	21.93644	0.00000
epc_labelB	0.14183	0.04133	3.43161	0.00061
epc_labelC	0.18429	0.03983	4.62667	0.00000
epc_labelD	0.21968	0.04390	5.00445	0.00000
epc_labelE	0.40952	0.04679	8.75243	0.00000
epc_labelF	0.38487	0.04937	7.79593	0.00000
building_era1970-2000	-0.00471	0.00934	-0.50470	0.61382
building_eraPost-2000	-0.00783	0.01505	-0.52033	0.60288
income_euro	0.00000	0.00000	5.64324	0.00000
occupancy_count	0.15091	0.00683	22.08676	0.00000
I(occupancy_count^2)	-0.00016	0.00071	-0.22363	0.82306
dist_to_brussels	-0.00011	0.00007	-1.55002	0.12127
roof_colorBlack	0.00328	0.00591	0.55537	0.57869
roof_colorOther	-0.00043	0.00600	-0.07123	0.94322
pet_ownership1	-0.00471	0.00501	-0.93995	0.34734
income_euro:occupancy_count	0.00000	0.00000	-0.04026	0.96789
sq_meters:income_euro	0.00000	0.00000	-8.15968	0.00000
epc_labelB:income_euro	0.00000	0.00000	-0.00914	0.99271
epc_labelC:income_euro	0.00000	0.00000	0.85972	0.39003

	Estimate	Std. Error	t value	p-value
epc_labelD:income_euro	0.00000	0.00000	1.33018	0.18358
epc_labelE:income_euro	0.00000	0.00000	0.17374	0.86209
epc_labelF:income_euro	0.00000	0.00000	0.24236	0.80852
sq_meters:epc_labelB	-0.00043	0.00031	-1.37878	0.16809
sq_meters:epc_labelC	-0.00027	0.00029	-0.94066	0.34697
sq_meters:epc_labelD	-0.00028	0.00032	-0.85880	0.39054
sq_meters:epc_labelE	-0.00069	0.00035	-1.99745	0.04589
sq_meters:epc_labelF	-0.00007	0.00036	-0.19128	0.84833
epc_labelB:occupancy_count	0.00302	0.00393	0.76904	0.44194
epc_labelC:occupancy_count	-0.00128	0.00375	-0.34103	0.73311
epc_labelD:occupancy_count	0.00662	0.00409	1.61845	0.10569
epc_labelE:occupancy_count	0.00307	0.00428	0.71825	0.47267
epc_labelF:occupancy_count	0.00396	0.00486	0.81575	0.41472

Table 7: Model comparison (initial vs. final) using AIC, BIC, and R-squared

	Model	AIC	BIC	R2
initial_model	initial_model	39972.402	40170.105	0.883
initial_model.log	initial_model.log	-3391.273	-3193.570	0.895
final_model	final_model	-3347.589	-3283.627	0.891

6.9 Diagnostic plots

Model regressing row annual energy consumption

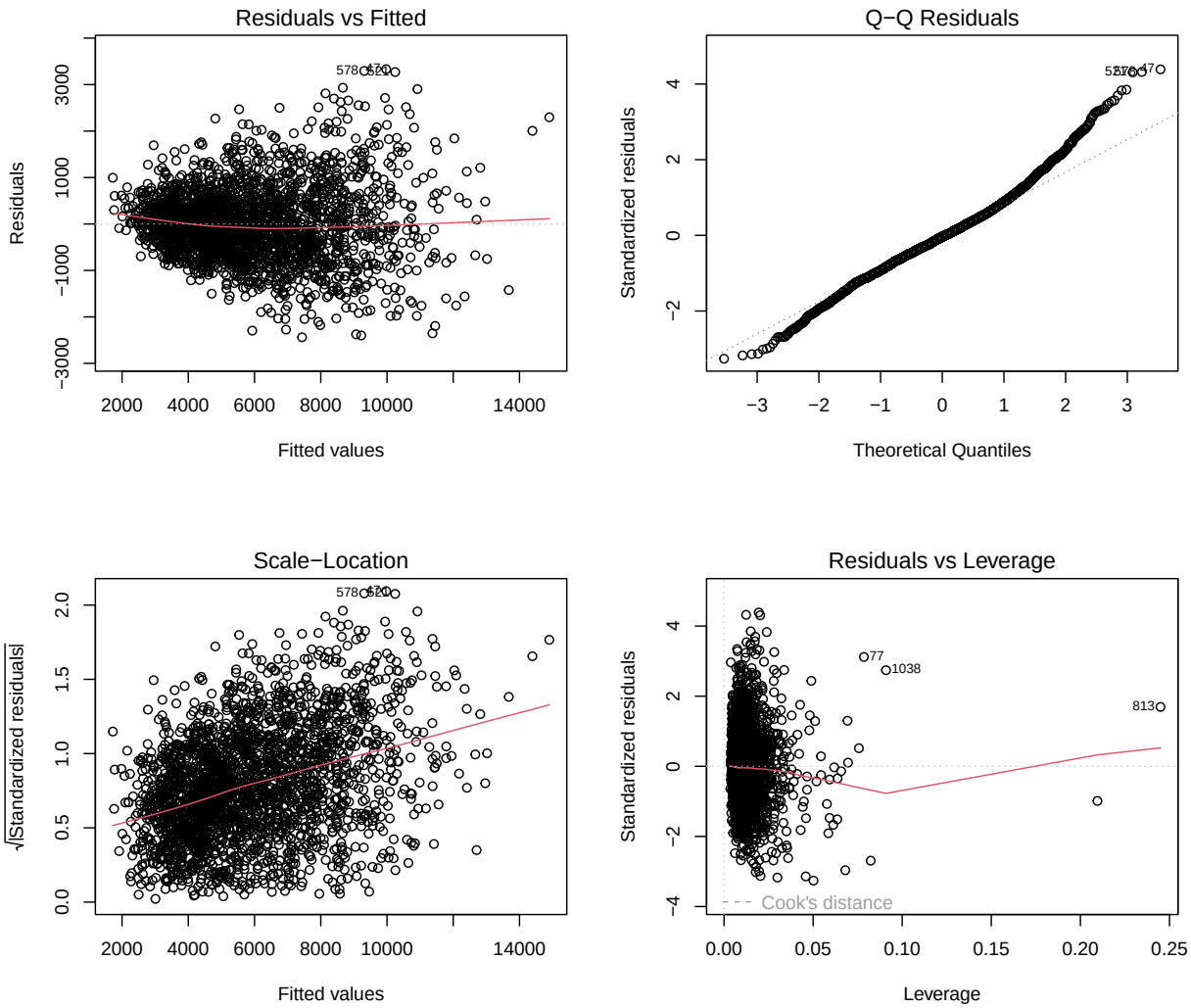


Figure 13: Diagnostic plots for the saturated regression model using annual energy consuption without log transformation.

Model regressing log-transformed annual energy consumption

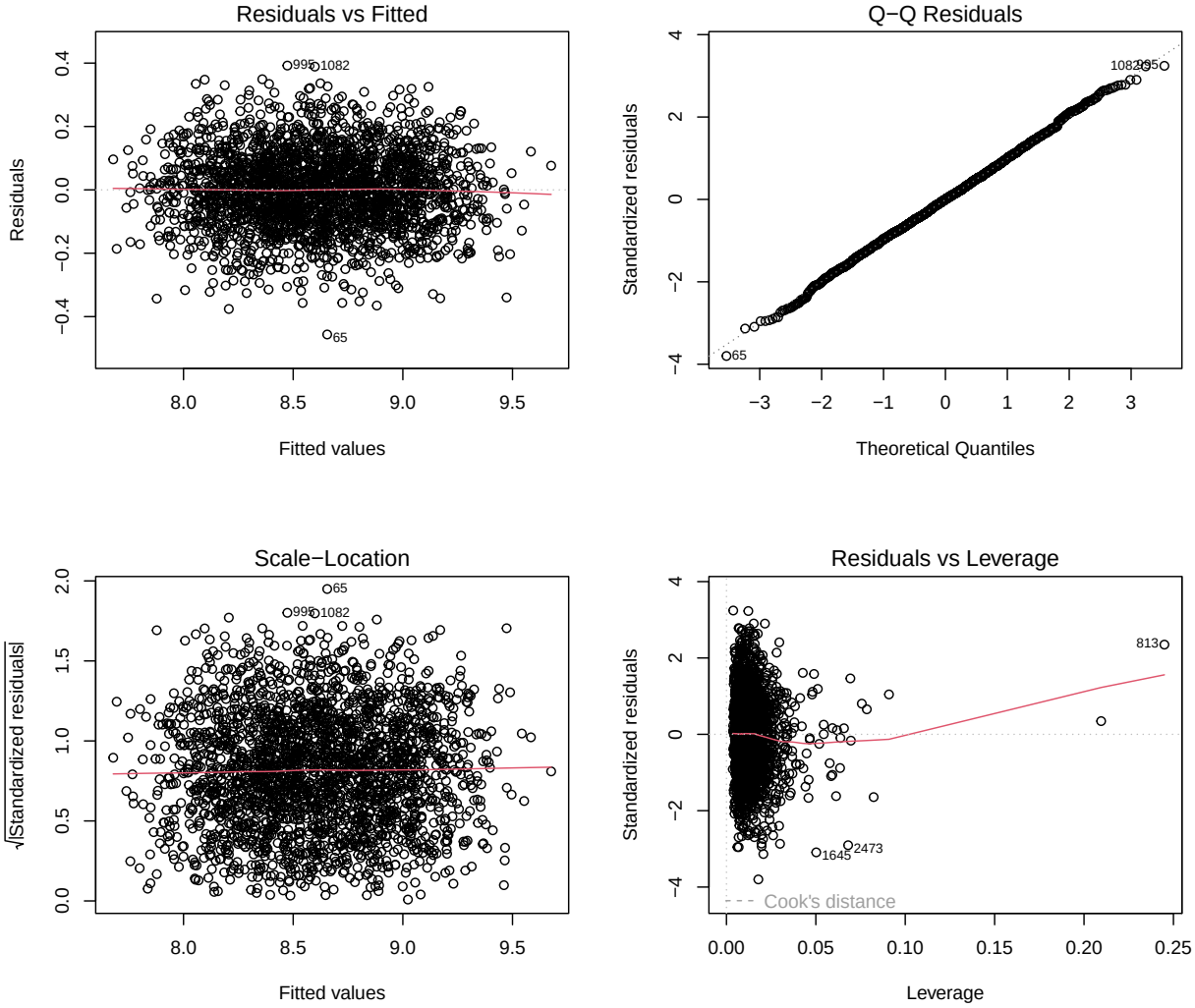


Figure 14: Diagnostic plots for the saturated regression model using log-transformed annual energy consuption.

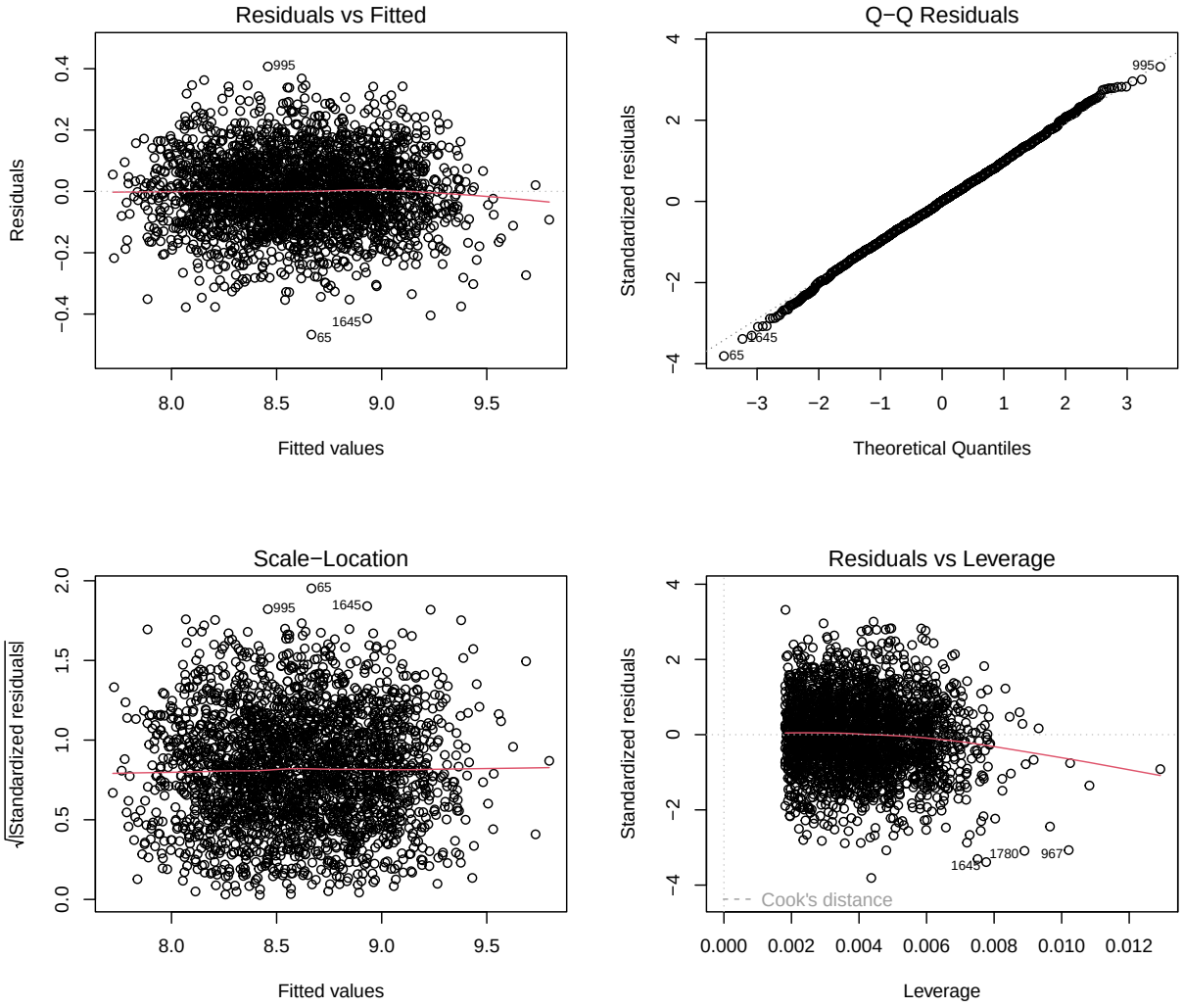


Figure 15: Diagnostic plots for the selected regression model using log-transformed annual energy consuption.

6.10 R codes

```
# creating the dataset
data.students <- read.table("data_students.txt",header=TRUE,sep=";")
df_clean <- data.students[-c(66, 36, 135), ] # removing obs per protocol
df_clean <- df_clean %>%
     filter(if_all(everything(), ~ !is.na(.))) # missing vals removed
head(df_clean[order(-df_clean$annual_kwh), ], 10)
df_clean <- df_clean %>%
    filter(annual_kwh <= 18000) # outliers in kWh removed

# Coding the variables
df_clean$epc_label <- factor(df_clean$epc_label)
df_clean$building_era <- factor(df_clean$building_era)
df_clean$roof_color <- factor(df_clean$roof_color)
df_clean$pet_ownership <- factor(df_clean$pet_ownership)

# Setting the reference levels
df_clean$building_era <- relevel(df_clean$building_era, ref = "Pre-1970")
df_clean$epc_label <- relevel(df_clean$epc_label, ref = "A")
df_clean$roof_color <- relevel(df_clean$roof_color, ref = "Red")
df_clean$pet_ownership <- relevel(df_clean$pet_ownership, ref = "0")

# testing the initial saturated model with all vars and interactions
full_model <- lm(
  annual_kwh ~ sq_meters +
    epc_label +
    building_era +
    income_euro +
    occupancy_count +
    I(occupancy_count^2) +
    dist_to_brussels +
    roof_color +
    pet_ownership +
    income_euro:occupancy_count +
    income_euro:sq_meters+
    income_euro:epc_label+
    epc_label:sq_meters +
    epc_label:occupancy_count,
  data = df_clean
)


n <- nrow(df_clean)

final_model <- step(
  full_model,
```

```r
  direction = "backward",
  k = log(n) # BIC criterion used
)


# visualise summary results (not displayed in the pdf)
summary(final_model, test=adjusted("BH"))
BIC(final_model)

# testing the initial saturated model with all vars and interactions but log_kWh
full_model.log <- lm(
  log(annual_kwh) ~ sq_meters +
    epc_label +
    building_era +
    income_euro +
    occupancy_count +
    I(occupancy_count^2) +
    dist_to_brussels +
    roof_color +
    pet_ownership +
    income_euro:occupancy_count +
    income_euro:sq_meters+
    income_euro:epc_label+
    epc_label:sq_meters +
    epc_label:occupancy_count,
  data = df_clean
)

n <- nrow(df_clean)

final_model.log <- step(
  full_model.log,
  direction = "backward",
  k = log(n)
)

# visualise summary results (not displayed in the pdf)
summary(final_model.log, test=adjusted("BH"))
BIC(final_model.log)

# Forcing building era into the model as a confounder

final_model.log_buil.era <- lm(
  log(annual_kwh) ~ sq_meters +
    epc_label +
    building_era+
    occupancy_count,
```

```r
  data = df_clean
)

# visualise summary results (not displayed in the pdf)
summary(final_model.log_buil.era, test=adjusted("BH"))
BIC(final_model.log_buil.era)

# CREATE A SUMMARY TABLE WITH REG RESULTS

# extract coefficients
coef_table <- summary(final_model.log_buil.era)$coefficients
ci <- confint(final_model.log_buil.era)

# Combine results
coef_table <- cbind(coef_table, ci)

# Convert to percentage effect on kWh
perc_effect <- (exp(coef_table[, "Estimate"]) - 1) * 100

# Transform CI
perc_ci_low  <- (exp(coef_table[, "2.5 %"]) - 1) * 100
perc_ci_high <- (exp(coef_table[, "97.5 %"]) - 1) * 100

# Keep selected columns (dropped beta CI and t statistics)
coef_table <- cbind(
  coef_table[, c("Estimate", "Std. Error", "Pr(>|t|)")],
  perc_effect, perc_ci_low, perc_ci_high
)

# Rename columns
colnames(coef_table) <- c(
  "Beta", "Std. Error", "p-value",
  "% Effect", "CI 2.5%", "CI 97.5%"
)

# Rename long row names
rownames(coef_table)[rownames(coef_table) ==
                       "building_era1970-2000"] <- "Built 1970-2000"
rownames(coef_table)[rownames(coef_table)
                     == "building_eraPost-2000"] <- "Built post-2000"

kable(coef_table,
      digits = 3,
      caption = "Regression results for the final model (p-values are adjusted
      using the Benjamini-Hochberg procedure)")

# show diagnostic plots for the final model
```

```r
par(mfrow = c(2,2))
plot(final_model.log_buil.era)
par(mfrow = c(1,1))

# Create summary tables for categorical variables
cat_summary <- df_clean %>%
  select(where(is.factor)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value") %>%
  group_by(variable, value) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(variable) %>%
  mutate(p = round(100 * n / sum(n), 1))

knitr::kable(cat_summary, caption = "Categorical variables (n, %)")

# Create summary tables for continuous variables
summary_table <- df_clean %>%
  summarise(
    across(
      c(annual_kwh, sq_meters, dist_to_brussels, income_euro),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Median = ~median(.x, na.rm = TRUE),
        IQR = ~IQR(.x, na.rm = TRUE),
        Min = ~min(.x, na.rm = TRUE),
        Max = ~max(.x, na.rm = TRUE)
      )
    )
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = c("Variable", "Statistic"),
    names_pattern = "(.+)_(Mean|SD|Median|IQR|Min|Max)$"
  ) %>%
  pivot_wider(names_from = Variable, values_from = value)

kable(summary_table,
      digits = 2,
      caption = "Summary Statistics for Continuous Variables")

missing_table <- data.students %>%
  summarise(across(everything(),
    ~ sum(is.na(.))
  )) %>%
  pivot_longer(everything(),
               names_to = "Variable",
```

```r
               values_to = "Missing_n") %>%
  mutate(
    Missing_pct = round(100 * Missing_n / nrow(df_clean), 2)
  ) %>%
  arrange(desc(Missing_pct))

knitr::kable(missing_table, caption = "Missing data summary")

# Box plots

layout(matrix(1:4, 2, 2))

boxplot(df_clean$annual_kwh,
        main = "Annual Energy Consumption",
        ylab = "annual_kwh")

boxplot(df_clean$sq_meters,
        main = "Floor Area",
        ylab = "sq_meters")

boxplot(df_clean$dist_to_brussels,
        main = "Distance to Brussels",
        ylab = "dist_to_brussels")

boxplot(df_clean$income_euro,
        main = "Income",
        ylab = "income_euro")

# QQ Plots

layout(matrix(1:4, 2, 2))


qqnorm(df_clean$annual_kwh,
       main = "QQ Plot: Annual Energy Consumption")
qqline(df_clean$annual_kwh)

qqnorm(df_clean$sq_meters,
       main = "QQ Plot: Floor Area")
qqline(df_clean$sq_meters)

qqnorm(df_clean$dist_to_brussels,
       main = "QQ Plot: Distance to Brussels")
qqline(df_clean$dist_to_brussels)

qqnorm(df_clean$income_euro,
       main = "QQ Plot: Income")
```

```r
qqline(df_clean$income_euro)

# QQ Plots for log_kWh

layout(matrix(1:4, 2, 2))

qqnorm(log(df_clean$annual_kwh),
       main = "QQ Plot: Log Annual Energy Consumption")
qqline(log(df_clean$annual_kwh))

qqnorm(log(df_clean$sq_meters),
       main = "QQ Plot: Log Floor Area")
qqline(log(df_clean$sq_meters))

qqnorm(log(df_clean$dist_to_brussels),
       main = "QQ Plot: Log Distance to Brussels")
qqline(log(df_clean$dist_to_brussels))

qqnorm(log(df_clean$income_euro),
       main = "QQ Plot: Log Income")
qqline(log(df_clean$income_euro))


# Box plot eergy consumption vs cat variables

par(mfrow = c(2,1))

boxplot(annual_kwh ~ epc_label,
        data = df_clean,
        main = "Annual Energy Consumption by EPC Label",
        ylab = "annual_kwh",
        xlab = "epc_label")

boxplot(annual_kwh ~ occupancy_count,
        data = df_clean,
        main = "Annual Energy Consumption by Occupancy Counts",
        ylab = "annual_kwh",
        xlab = "Occupancy Counts")

par(mfrow = c(2,1))

boxplot(annual_kwh ~ building_era,
        data = df_clean,
        main = "Annual Energy Consumption by Building Era",
        ylab = "annual_kwh",
        xlab = "building_era")
```

```r
boxplot(annual_kwh ~ pet_ownership,
        data = df_clean,
        main = "Annual Energy Consumption by Pet Ownership",
        ylab = "annual_kwh",
        xlab = "Pet_Ownership")

par(mfrow = c(1,1))

boxplot(annual_kwh ~ roof_color,
        data = df_clean,
        main = "Annual Energy Consumption by Roof Color",
        ylab = "annual_kwh",
        xlab = "Roof Color")


# Bar plot of categorical vars against each-other

par(mfrow = c(2,1))

df_clean$building_era <- as.factor(df_clean$building_era)
df_clean$epc_label <- as.factor(df_clean$epc_label)
df_clean$occupancy_count <- as.factor(df_clean$occupancy_count)

cols2 <- gray.colors(length(levels(df_clean$epc_label)))

barplot(table(df_clean$epc_label, df_clean$building_era),
        beside = TRUE,
        main = "ECP label by Building Era",
        ylab = "count",
        xlab = "Building Era")

legend("topright",
       inset = c(-0.05, 0),   # move right
       legend = levels(df_clean$epc_label),
       fill = cols2,
       cex = 0.8,
       bty = "n",
       xpd = TRUE)

cols3 <- gray.colors(length(levels(df_clean$building_era)))

barplot(table(df_clean$occupancy_count, df_clean$building_era),
        beside = TRUE,
        col = cols3,
        main = "Occupancy Count Distribution by Building Era",
        ylab = "Count",
        xlab = "Building Era")
```

```r
legend("topright",
       inset = c(-0.05, 0),   # move right
       legend = levels(df_clean$occupancy_count),
       fill = cols3,
       cex = 0.8,
       bty = "n",
       xpd = TRUE)


par(mfrow = c(1,1))


# Divide income by quantile
df_clean$income_q <- cut(
  df_clean$income,
  breaks = quantile(df_clean$income, probs = seq(0, 1, 0.2), na.rm = TRUE),
  include.lowest = TRUE,
  labels = c("Q1", "Q2", "Q3", "Q4", "Q5")
)

# Table EPC labels distribution by income quintile
tab <- table(df_clean$income_q, df_clean$epc_label)
cols <- gray.colors(length(levels(df_clean$epc_label)))


bp <- barplot(t(tab),
              beside = TRUE,
              col = cols,
              main = "EPC Label Distribution by Income",
              ylab = "Count",
              xlab = "Income Quintile",
              names.arg = levels(df_clean$income_q))

legend("topright",
       inset = c(-0.1, 0),   # move right
       legend = levels(df_clean$epc_label),
       fill = cols,
       cex = 0.8,
       bty = "n",
       xpd = TRUE)

# Divide sq_meters by quantile
df_clean$sq_meters_q <- cut(
  df_clean$sq_meters,
  breaks = quantile(df_clean$sq_meters, probs = seq(0, 1, 0.2), na.rm = TRUE),
  include.lowest = TRUE,
  labels = c("Q1", "Q2", "Q3", "Q4", "Q5")
```

```r
)


# Visualize sq_meters by occupancy_count
df_clean$occupancy_count <- as.factor(unlist(df_clean$occupancy_count))
boxplot(sq_meters ~ occupancy_count,
        data = df_clean,
        main = "Dwelling Size by Occupants",
        xlab = "Occupancy count",
        ylab = "Square meters")

# Table mean Annual Energy Consumption by Size and Income Quintiles
tab_mean <- tapply(df_clean$annual_kwh,
                   list(Size = df_clean$sq_meters_q,
                        Income = df_clean$income_q),
                   mean,
                   na.rm = TRUE)
kable(tab_mean,
      caption = "Mean Annual Energy Consumption by Size and Income Quintiles",
      digits = 0,
      row.names = TRUE) %>%
  kableExtra::add_header_above(c(" " = 1, "Income Quintiles" = ncol(tab_mean)))



# Correlation plots of each continuous variable with eachother

par(mfrow = c(1,1))  # reset plotting layout

vars <- df_clean[, c("annual_kwh", "sq_meters", "dist_to_brussels",
                     "income_euro")]
cor_matrix <- cor(vars, use = "complete.obs")
corrplot(cor_matrix,
         method = "circle",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black",
         tl.srt = 45)


# Create DAG for the causal relationship among predictors and aoutcome

# DAG 1: Confounder
dag_confounder <- dagitty("
dag {
  building_era -> epc_label
  building_era -> annual_kwh
```

```
  epc_label -> annual_kwh
}
")

p1 <- ggdag(dag_confounder, text = TRUE, use_labels = "name") +
  theme_dag() +
  ggtitle("Building Era as a Confounder")

# DAG 2: Mediator
dag_mediator <- dagitty("
dag {
  income -> sq_meters
  sq_meters -> annual_kwh
}
")

p2 <- ggdag(dag_mediator, text = TRUE, use_labels = "name") +
  theme_dag() +
  ggtitle("Square_meters as a Mediator")

# Combine plots
p1 + p2

# Display reg results for the saturated model

coef_table <- summary(full_model.log)$coefficients
colnames(coef_table) <- c("Estimate", "Std. Error", "t value", "p-value")

kable(coef_table,
      digits = 5,
      caption = "Regression results for the saturated model using l
      og-transformed outcome (p-values are adjusted using the
      Benjamini-Hochberg procedure).",
      booktabs = TRUE,
      escape = TRUE)


# Compare BIC, AIC and R2 for each model tested

models <- list(
  initial_model = full_model,
  initial_model.log = full_model.log,
  final_model = final_model.log_buil.era
)

data.frame(
  Model = names(models),
```

```r
  AIC = sapply(models, AIC),
  BIC = sapply(models, BIC),
  R2 = sapply(models, function(m) summary(m)$r.squared)
)

model_comp <- data.frame(
  Model = names(models),
  AIC = sapply(models, AIC),
  BIC = sapply(models, BIC),
  R2 = sapply(models, function(m) summary(m)$r.squared)
)

kable(model_comp, digits = 3, caption = "Model comparison (initial vs. final)
      using AIC, BIC, and R-squared")

# Diagnostic plots for each model
par(mfrow = c(2,2))
plot(full_model)
plot(full_model.log)
plot(final_model.log_buil.era)
par(mfrow = c(1,1))
```

References

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McLoughlin, F., Duffy, A., & Conlon, M. (2013). Determinants of residential electricity consumption: Using smart meter data to examine the effect of climate, building characteristics, appliance stock, and occupants' behavior. *Energy*, *55*, 184–194.

Santin, O. G., Itard, L., & Visscher, H. (2009). The effect of occupancy and building characteristics on energy use for space and water heating in dutch residential stock. *Energy and Buildings*, *41*(11), 1223–1232.